

From Servers & Switches to Universal Chiplet Nodes

Session Proposal.

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Introduction

In graph theory terms, a datacenter is simply a set of nodes connected by edges. Yet historically, we have enforced a rigid distinction between *servers* (application hosts) and *switches/routers* (network appliances). This proposal traces how that dichotomy arose, and then explores opportunities to collapse it into a model of *universal nodes*.

A Short History of Why Switches/Routers and Servers Diverged

1. **Early packet networks:** ARPANET's Interface Message Processors (IMPs) were dedicated forwarders, distinct from hosts

Architectural Opportunities to Collapse the Boundary

Universal Node Concept

Every node combines:

- (a) A programmable line-rate packet pipeline (P4-enabled ASIC).
- (b) A DPU/IPU for IO, storage, and security.
- (c) CPU/GPU cores for applications.
- (d) Fast local memory (DDR5/HBM).

Depending on assignment, a node behaves like a switch, a server, or both.

Rack-Scale Design

- Each rack unit is a universal node.
- The controller allocates both compute cycles and forwarding entries.
- No discrete ToR/server divide.

Pod-Scale Design

- Clos/fat-tree built entirely from universal nodes.
- Leaf/spine are logical roles, not hardware SKUs.
- Central scheduler allocates tasks and flows jointly.

Unified Chiplet Architecture (UCA)

Why SONiC is Insufficient

- SONiC is a disaggregated network OS, not a general-purpose scheduler.
- It assumes servers and switches are distinct entities.
- It cannot natively place compute tasks alongside packet pipelines.

What UCA Provides

- UCA (Unified Chiplet Architecture) spans chiplet-based CPUs, GPUs, DPUs, and programmable pipelines.
- Scheduling primitive: treats forwarding entries and compute cycles as fungible resources.
- Northbound: Kubernetes-like API for jobs and flows.
- Southbound: P4Runtime, DPDK/eBPF, gRPC/OCI.

Deploying a microservice and deploying a packet-forwarding function become operationally identical. Unlike SONiC, UCA is designed from the start to be chiplet-native.

Chiplets and the Datacenter

Motivation for Chiplets

- Moore's Law scaling pressures have made monolithic dies increasingly costly.
- Chiplets allow mixing of process technologies (e.g., logic vs. memory).
- Standards such as UCle provide interoperability across vendors.

Chiplet Summit Insights

The Chiplet Summit (2023–2025) emphasized that:

- Chiplets enable heterogeneous integration of compute, memory, and networking

Implications for UCA

- Datacenters can be constructed from uniform chiplet-based universal nodes.
- The line between “server,” “switch,” and “accelerator” disappears both at rack-scale and at package-scale.
- UCA unifies orchestration across these layers.

*Workload Case Studies**Machine Learning Training*

- Traditional: gradient aggregation at servers; fabric carries raw tensors.
- UCA: in-fabric aggregation at line rate (as in SHARP and SwitchML).
- Gains: lower p99.9 latency, reduced GPU idle time.

Distributed Storage

- Traditional: replication and erasure coding handled in storage daemons.
- UCA: parity offloaded to packet pipelines.
- Gains: faster recovery, lower server overhead.

RPC-heavy Services

- Traditional: TCP retries, head-of-line blocking, load balancers.
- UCA: ingress caching and flowlet scheduling at fabric scale.
- Gains: improved tail latency, fewer retries.

Experiment Matrix

Metric	Traditional DC	UCA Datacenter	Proof Workload
Tail latency (p99.9)	Host aggregation	In-fabric reduce	ML training
Energy/bit	NIC → switch → server	In-fabric ops	Key/Value store
Upgrade velocity	Two stacks (NOS + OS)	Unified pipeline	Canary deployment
Failure isolation	Box boundaries	Node boundaries	Node/link failure
Dev agility	Vendor firmware cycle	P4/UCA push	Telemetry parser

Incremental vs Radical Path

Incremental Path

- Introduce P4-capable ToR switches.
- Deploy DPUs in servers.
- Gradually evolve SONiC into UCA.

Radical Path

- Deploy only UCA-enabled universal nodes from day one.
- Replace SONiC outright with UCA.
- Market edge: single SKU, chiplet-native architecture, unified ops.

Conclusion

The server/switch split arose from historical divides. With programmable pipelines, chiplets, and DPUs, we can now design datacenters as graphs of *universal chiplet-based nodes*. This simplifies hardware, unifies operations, and disrupts the current model by eliminating the boundary between server and switch.

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